

SIMATS ENGINEERING
ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I M. AJAY / 192421031 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Ajay . M

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud.
The system must handle millions of concurrent users with minimal buffering.
Design a cloud architecture using scalability and caching techniques.
Explain how load balancing and auto-scaling can ensure smooth streaming.
Discuss cost optimization strategies for such high-demand workloads.
2. A fintech startup is deploying applications using containers across multiple cloud providers.
They face challenges in maintaining consistency and managing deployments.
Propose a solution using container orchestration and multi-cloud strategies.
Explain how service discovery and scaling can be handled efficiently.
Highlight the benefits and risks of a multi-cloud deployment model.
3. An e-learning platform experiences data loss due to accidental deletion in the cloud.
The organization needs a robust backup and disaster recovery strategy.
Design a solution using cloud-native backup, replication, and versioning.

- Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.
Provide steps to ensure business continuity during failures.
4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.
 5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Issues / Concepts	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Conclusion / Recommendation	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Summary	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

Cloud Architecture:

- * Use cloud servers with load balancer, Auto scaling, Content Delivery Network (CDN).
- * Cache frequently accessed videos using CDN to reduce buffering.

Load Balancing & Auto scaling:

- * Load balancer distributes user requests evenly across servers.
- * Auto scaling automatically adds or removes servers based on traffic.

Cost Optimization:

- * Use pay-as-you-go pricing.
- * Enable auto scaling to avoid idle resources.
- * Store old ~~retrieved~~ videos in low-cost storage.

This architecture provides high performance, low latency and cost-efficient streaming.

Solution:

Use Docker containers with Kubernetes for orchestration.
Deploy applications across multiple cloud providers.

Service Discovery & Scaling:

- * Kubernetes automatically discovers services.
- * Horizontal pod Autoscale increases or decreases container instances based on demand.

Benefits:

- * High availability.
- * Vendor lock-in is reduced.
- * Better disaster recovery.

Risks:

- * Complex management
- * Higher networking and monitoring costs.

Solution:

- * Enable automatic cloud backups
- * Use data replication across multiple regions.
- * Enable object versioning to restore deleted files.

RTO & RPO:

- * RTO (Recovery Time Objective): Time required to resume service

- * RPO (Recovery point Objective): Maximum acceptable loss.

Business Continuity steps:

- *. schedule regular backups
- *. test disaster recovery plans
- *. monitor backup status continuously.

4.

1: Hybrid Architecture:

- *. store sensitive data on-premises.
- *. store non-sensitive application in the public cloud.
- *. Connect both environments securely using VPN or dedicated links.

Identity Management & Encryption:

- *. use single sign-on (SSO) and Multi-Factor Authentication (MFA).
- *. Encrypt data both at rest and during transmission.

challenges:

- *. Managing Multiple environments
- *. Ensuring compliance
- *. Network Latency and security management.

Multi-Region Deployment:

- * Deploy the application in multiple cloud regions.
- * Configure automatic failover if one region fails.

Replication & Failover:

- * Replicate databases across regions
- * Use health checks and load balancers to redirect traffic automatically.

Monitoring & Alerting:

- * Monitor CPU, memory, storage, and network.
- * Configure alerts for failures and performance issues
- * Use cloud monitoring tools for proactive detection

Conclusion: Multi-region deployment ensures high availability, fault tolerance, and uninterrupted service.